## ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** glow dimly | 4.5 V is less than the rated 6 V. |
| 2 | **(b)** test with a tester; add salt to water to make it conduct | Dissolved salts supply ions. |
| 3 | **(c)** a conductor | Current passes through it. |
| 4 | **(c)** likely fuse due to 7.5 V | Exceeds rating. |
| 5 | **(a)** tap water has dissolved salts | Dissolved ions conduct. |
| 6 | **(b)** an insulator | Current does not pass. |
| 7 | **(c)** series | Single path. |
| 8 | **(a)** copper conducts better | Brightness suggests current. |
| 9 | **(a)** both conduct, copper better | Brightness shows current. |
| 10 | **(a)** a single path for the current | The loop is broken. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Tungsten has a very high melting point (1); Copper would melt before glowing (1). | 1+1 |
| 12 | Metals are good conductors (1); They carry current to the appliance (1). | 1+1 |
| 13 | Prevent touching live parts (1); Avoid shocks (1). | 1+1 |
| 14 | Nichrome has high resistance (1); It does not oxidise at high temperatures (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | 6 x 1.5 = 9 V (1); Much higher than 3 V (1); Bulb may fuse (1). | 1+1+1 |
| 16 | It needs a large current (1); Thin wiring may overheat (1); Risk of fire (1). | 1+1+1 |
| 17 | Switch off the supply (1); Dry hands (1); Use proper bulb (1). | 1+1+1 |
| | **OR:** Thin wire heats more (1); Melts first (1); Protects the circuit (1). | 1+1+1 |
| 18 | Iron would melt or burn (1); Tungsten has a high melting point (1); Bulb has inert gas (1). | 1+1+1 |
| | **OR:** One path (1); Same current (1); If one fails all stop (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | All bulbs go off (1); Only one path (1); Test each bulb with a tester (1); Replace the faulty one (1); Use a fresh bulb for check (1). | 1+1+1+1+1 |
| | **OR:** Marks show quality (1); Safer products (1); Fuse rating matches use (1); Do not use higher rating (1); Check before buying (1). | 1+1+1+1+1 |
| 20 | Switch off unused devices (1); Use LED bulbs (1); Use daylight (1); Do not overload (1); Service appliances (1). | 1+1+1+1+1 |
| | **OR:** Overloading (1); Short circuits (1); Damaged wires (1); Use ISI products (1); Use fuse or MCB (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** the nail is coated with paint or rust | Surface layer blocks contact. |
| 2 | **(d)** goes off | The loop is broken. |
| 3 | **(c)** its melting point is lower than that of the connecting wires | Low melting. |
| 4 | **(c)** touch it | Touching is unsafe. |
| 5 | **(b)** the reversed cell reduces the total voltage | Opposed cells cancel partly. |
| 6 | **(d)** the lemon juice conducts | Ions in the juice. |
| 7 | **(b)** to avoid contact and shock | Safety. |
| 8 | **(c)** a broken connecting wire | Open path. |
| 9 | **(d)** the bulb has inert gas or vacuum | No oxygen. |
| 10 | **(d)** exposed wire can shock | Insulation damage. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Cells in series add their voltages (1); Higher voltage drives more current through the filament (1). | 1+1 |
| 12 | To carry current safely to the earth in a fault (1); Prevent shock (1). | 1+1 |
| 13 | 15 A fuse will not protect thin wires (1); They may overheat (1). | 1+1 |
| 14 | The appliance is not live when switched off (1); It reduces the risk of shock (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | 9 / 1.5 = 6 (1); Six cells in series (1); + to - connections (1). | 1+1+1 |
| 16 | Switch off appliances (1); Replace the fuse with the same rating (1); Avoid overloading (1). | 1+1+1 |
| 17 | Chemicals may leak (1); Damage to the toy (1); Cell is saved (1). | 1+1+1 |
|  | **OR:** Opposing cells reduce voltage (1); Little or no current (1); Filament does not heat (1). | 1+1+1 |
| 18 | Use less energy (1); Less fuel burnt (1); Fewer emissions (1). | 1+1+1 |
|  | **OR:** Saves electrical energy (1); Reduces bills (1); Saves resources (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Same cells, wires of different length (1); Hold match heads or wax on wire (1); Observe time to heat (1); Compare (1); Safe handling (1). | 1+1+1+1+1 |
|  | **OR:** Heating plate: metal alloy (1); Body: plastic insulator (1); Reason: shock safety (1); Light weight (1); Heat resistant plastic (1). | 1+1+1+1+1 |
| 20 | Insulated wiring (1); Fuse and MCB (1); Earthing (1); Dry hands (1); Switch off and call help (1). | 1+1+1+1+1 |
|  | **OR:** Light at night (1); Machines run (1); Communication (1); Rules prevent accidents (1); Careful use (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Electricity: Circuits and their Components** | Class 7 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** iron nail, rubber, aluminium foil | Conductor, insulator, conductor. |
| 2 | **(a)** heating effect | Wire melts by heat. |
| 3 | **(d)** a sealed metal sheath with insulating powder | Safe design. |
| 4 | **(c)** it can carry more current without overheating | Thick wires heat less. |
| 5 | **(c)** a long wire fits in a small space | Compact design. |
| 6 | **(a)** switching to LED bulbs | Efficient. |
| 7 | **(d)** water conducts and can give a shock | Safety. |
| 8 | **(a)** connecting many appliances to one socket | Large current. |
| 9 | **(c)** open | No complete path. |
| 10 | **(c)** replacing a blown fuse of the same rating after switching off the supply | Safe practice. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Current flows quickly through the circuit (1); All parts conduct (1). | 1+1 |
| 12 | Positive of one to negative of next (1); 3 x 1.5 V = 4.5 V (1). | 1+1 |
| 13 | Thin wires overheat (1); Safe current carrying (1). | 1+1 |
| 14 | Conduct and heat (1); Can be shaped as coils (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Large current (1); Wires heat (1); Insulation burns (1). | 1+1+1 |
| 16 | Distilled has no ions (1); Tap water has dissolved salts (1); So it conducts (1). | 1+1+1 |
| 17 | Lighter (1); Cheaper (1); Good conductor (1). | 1+1+1 |
| | **OR:** Dip tester ends (1); Bulb glows for conducting liquids (1); Lemon and salt solution conduct, sugar does not (1). | 1+1+1 |
| 18 | More current more heat (1); Heat increases with current (1); Fuse chosen accordingly (1). | 1+1+1 |
| | **OR:** Overheating (1); Damage to holder and wiring (1); Fire (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | 3 cells in series (1); Total 4.5 V (1); Bulb glows normally (1); 2 cells give 3 V (1); Bulb dim (1). | 1+1+1+1+1 |
| | **OR:** Used in iron, heater (1); Easy to control (1); Energy wasted in devices (1); Wire heating hazard (1); Use fuse (1). | 1+1+1+1+1 |
| 20 | Tester shows conduction (1); Metals light the bulb (1); Poor conductors give a weak current (1); Needs good contacts (1); Cannot measure resistance (1). | 1+1+1+1+1 |
| | **OR:** In series with the appliance (1); In the live wire (1); Current passes through it (1); Melts on overload (1); Appliance isolated (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.